HeartPro — Single-lead ECG

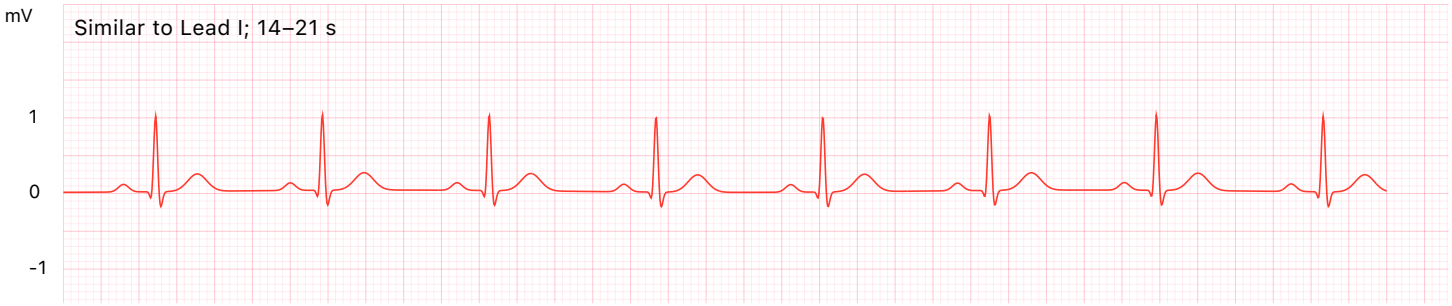
Friday, September 18, 2026 at 4:08 PM

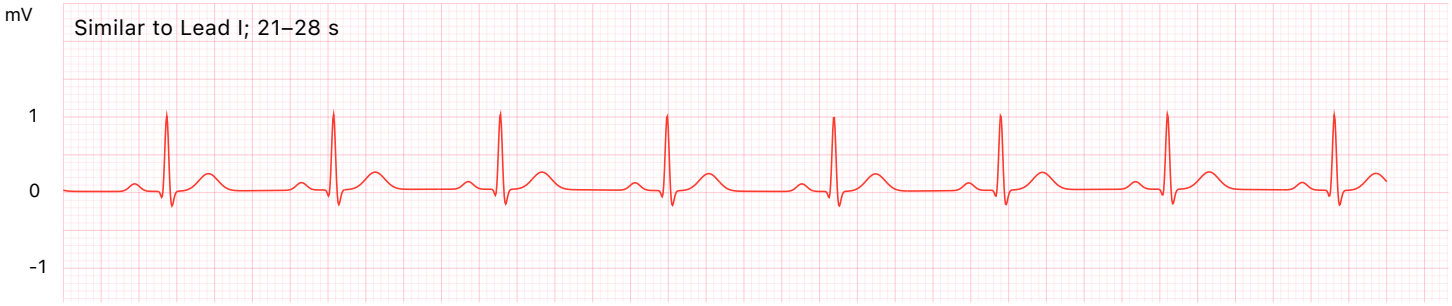
Synthetic demo: Apple Watch; 30 s; 68 bpm; Sinus rhythm

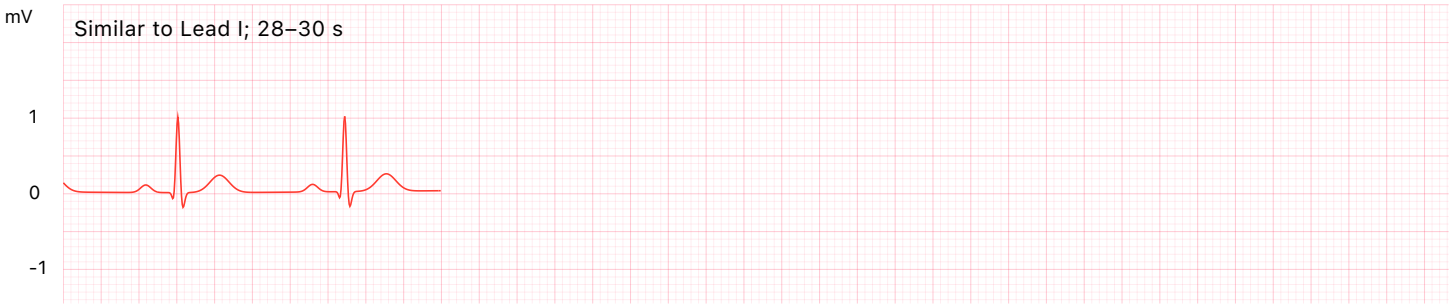
25 mm/s; 10 mm/mV; print at 100% actual size; page 1











For reference only, not for medical use. Cannot replace diagnosis or rule out a heart attack or myocardial ischemia.

Analysis version: 0.4.0-research; original ECG is read-only

HeartPro — Recording statistics

Usable recording time: 100%; 30 s

R peaks identified: 34; RR intervals: 33

Median RR: 883 ms

Overall interval variation (RR standard deviation): 1 ms

Successive interval changes (root mean square of RR differences): 2 ms

Normal beats have not been distinguished from other types, so these statistics cannot be treated as HRV.

Waveform changes marked: 0

The app estimates the positions and timing of ECG segments.

Analysis version: 0.4.0-research

Automatic estimates, not validated on real Watch data. For reference only, not for medical use.

HeartPro — Premature-beat candidates and rhythm

Beats reviewed: 32 of 34 identified

Rhythm reference score: 100 / 100

Reviewable beats: 32; flagged: 0. Each beat counts at most once.

Premature-beat candidates in this recording: 0

Automatic detection can miss events or flag them incorrectly. Review candidates against the original trace.

Rhythm reference score = 100 × (1 – flagged beats ÷ reviewable beats), rounded. Flags include premature-beat candidates, marked interval changes or local waveform changes. The pause after a candidate is not counted again. This describes the share of beats not flagged by these rules in this recording. It is not a heart-health score, and 100 does not mean the ECG is normal.

At least 20 seconds of clear signal, 80% signal coverage, 15 reviewable beats and 8 stable reference beats are required. At least 80% of non-edge beats must be reviewable. Poor signal does not reduce the score; missing PR or QT values and higher or lower HRV do not add or subtract points.

Compare with nearby beats: the beat must arrive over 15% and at least 80 ms early, with a longer following interval or a wider, changed QRS. Wide candidates require a local width of at least 120 ms, over 20% wider than the reference, and template correlation below 0.90. Narrow candidates require a width under 120 ms, correlation of at least 0.90 and a longer following interval. These research rules cannot by themselves confirm atrial or ventricular premature beats.

Rule version: rhythm-review-1

Automatic estimates, not validated on real Watch data. For reference only, not for medical use.

HeartPro — ECG segments

Synthetic demo; heartbeats identified: 34

These values are estimated automatically by the app. Accuracy has not been validated on real Apple Watch data, and the values cannot determine whether an ECG is normal.

Understanding the numbers

ms means milliseconds: 1,000 ms equals 1 second. mV means millivolts and describes the waveform's level relative to a reference baseline. Positive values are above it; negative values are below.

The median is the middle value when measurements are sorted from smallest to largest. It is not the average.

Min–max covers all measured values. The middle 80% range (P10–P90) excludes the lowest and highest 10% to show where values cluster. Neither is a normal range.

Counts are based on beats the app identifies; some beats may be missed. Beats without measurements are not replaced with zero or included in statistics.

P wave

The P wave reflects the atria's electrical activity and usually appears before the QRS complex. Here you can see its duration, peak height and whether it points up or down.

P-wave duration: Median 121 ms; min–max 109–133 ms; middle 80% 113–133 ms; Measurable beats: 33 of 34

P-wave amplitude: Median 0.10 mV; min–max 0.09–0.10 mV; middle 80% 0.09–0.10 mV; Measurable beats: 33 of 34

Why measurement isn't possible:

Occurrences: 1. The waveform around this beat is incomplete, or part of the signal is unclear.

A common adult reference for P-wave duration is less than 120 ms. Peak height also depends on the recording lead.

Measurement method: The start, end and peak of the P wave are estimated. If it is too small, its boundaries are unclear or it overlaps nearby waves, the affected values are left blank.

PR interval

The PR interval runs from the start of the P wave to the start of the QRS complex. It helps show how the electrical signal travels from the atria to the ventricles.

PR interval: Median 188 ms; min–max 184–191 ms; middle 80% 184–191 ms; Measurable beats: 33 of 34

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HeartPro — ECG segments

Why measurement isn't possible:

Occurrences: 1. The waveform around this beat is incomplete, or part of the signal is unclear.

A common adult reference is 120–200 ms. Age, heart rate and measurement method affect the result.

Measurement method: Uses the P-wave start and QRS start from the same beat. If either is unclear, no PR interval is calculated. PR includes the P wave itself and the following PR segment.

PR segment

This is the short segment between the end of the P wave and the start of the QRS complex. Here you can see its duration and its level relative to the reference baseline.

PR-segment duration: Median 66 ms; min–max 59–74 ms; middle 80% 59–73 ms; Measurable beats: 33 of 34

PR-segment level: Median 0.00 mV; min–max -0.01–0.01 mV; middle 80% -0.01–0.01 mV; Measurable beats: 33 of 34

Why measurement isn't possible:

Occurrences: 1. The waveform around this beat is incomplete, or part of the signal is unclear.

The PR segment is usually close to the reference baseline. Baseline placement and signal processing affect its measurement.

Measurement method: Measured from the P-wave end to QRS start. The level is the mean voltage of this segment relative to the local TP baseline. Positive values are above the baseline; negative values are below it.

QRS complex

The QRS complex is usually the tall, sharp group of waves in an ECG. It reflects the ventricles' electrical activity. Here you can see its total duration and the size of the Q, R and S components.

QRS duration: Median 98 ms; min–max 94–98 ms; middle 80% 94–98 ms; Measurable beats: 33 of 34

Q amplitude (estimated): Median -0.07 mV; min–max -0.08–-0.06 mV; middle 80% -0.07–-0.06 mV; Measurable beats: 33 of 34

Q duration (estimated): Median 20 ms; min–max 16–20 ms; middle 80% 16–20 ms; Measurable beats: 33 of 34

Main R-peak amplitude: Median 0.94 mV; min–max 0.92–0.95 mV; middle 80% 0.92–0.95 mV; Measurable beats: 33 of 34

QRS start to R peak: Median 43 ms; min–max 39–43 ms; middle 80% 39–43 ms; Measurable

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beats: 33 of 34

S amplitude (estimated): Median -0.18 mV; min–max -0.19–-0.17 mV; middle 80% -0.19–-0.17 mV; Measurable beats: 33 of 34

S duration (estimated): Median 31 ms; min–max 27–35 ms; middle 80% 28–35 ms; Measurable beats: 33 of 34

Why measurement isn't possible:

Occurrences: 1. The waveform around this beat is incomplete, or part of the signal is unclear.

A common adult reference for QRS duration is less than 120 ms. Q, R and S identification should also be checked against the original trace.

Measurement method: QRS start and end are estimated first. Opposite-direction waves before and after the main peak are candidates for Q and S. Values retain their original polarity. If the main peak points down, Q, R and S labels need manual review. A single lead cannot assess the electrical axis or R-wave progression across chest leads.

ST segment / J point

The ST segment lies between QRS and the T wave. The end of QRS is called the J point. Here you can see whether the waveform is above or below the reference baseline and which way it changes.

ST-segment duration: Median 59 ms; min–max 47–66 ms; middle 80% 47–66 ms; Measurable beats: 33 of 34

J-point level: Median -0.01 mV; min–max -0.02–0.01 mV; middle 80% -0.02–0.00 mV; Measurable beats: 33 of 34

Level 60 ms after J: Median 0.02 mV; min–max 0.01–0.02 mV; middle 80% 0.01–0.02 mV; Measurable beats: 9 of 34

Level 80 ms after J: Unable to measure; Measurable beats: 0 of 34

ST voltage change rate: Unable to measure; Measurable beats: 0 of 34

Why measurement isn't possible:

Occurrences: 33. The T-wave start is unclear, so the ST measurement window cannot be determined.

Occurrences: 33. This position is too close to the T wave to measure ST separately.

Occurrences: 1. The waveform around this beat is incomplete, or part of the signal is unclear.

Contact, posture and signal processing can all change the ST segment's appearance. A Watch ECG cannot use this to diagnose or rule out myocardial ischemia.

Measurement method: The QRS end estimates the J point. Voltage is measured 60 and 80 ms later. The change rate comes from the voltage difference between these positions: positive means

HeartPro — ECG segments

upward and negative downward. Positions too close to the T wave are not used for ST measurements.

T wave

The T wave reflects electrical recovery of the ventricles. Here you can see its duration, peak size and the shape of its rising and falling parts.

T-wave duration: Median 219 ms; min–max 195–254 ms; middle 80% 199–246 ms; Measurable beats: 33 of 34

T-wave amplitude: Median 0.23 mV; min–max 0.22–0.24 mV; middle 80% 0.22–0.24 mV; Measurable beats: 33 of 34

T-wave duration ratio: Median 0.97 ×; min–max 0.79–1.07 ×; middle 80% 0.90–1.04 ×; Measurable beats: 33 of 34

Why measurement isn't possible:

Occurrences: 1. The waveform around this beat is incomplete, or part of the signal is unclear.

T-wave direction and size vary by lead. A single value cannot determine whether it is normal.

Measurement method: The T-wave start, peak and end are found separately. The duration ratio is start-to-peak time divided by peak-to-end time. It is near 1 when the two parts have similar lengths. Multiple peaks require checking the original trace.

QT interval

The QT interval runs from the start of QRS to the end of the T wave. It covers the ventricles' electrical activity from activation through recovery.

QT interval: Median 371 ms; min–max 359–395 ms; middle 80% 359–394 ms; Measurable beats: 33 of 34

Why measurement isn't possible:

Occurrences: 1. The waveform around this beat is incomplete, or part of the signal is unclear.

QT changes with heart rate, so one fixed range cannot be used to assess all records.

Measurement method: Each beat is measured separately, from its QRS start to its T-wave end, before results are summarized. If the T-wave end is unclear, QT is not calculated.

QTc (corrected QT)

Heart rate affects QT. QTc adjusts QT using the heartbeat interval to make comparisons easier. Two common formulas are shown below; their results may differ slightly.

QTc (Fridericia): Median 387 ms; min–max 375–412 ms; middle 80% 375–407 ms; Measurable

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beats: 31 of 34

QTc (Bazett): Median 395 ms; min–max 382–421 ms; middle 80% 382–416 ms; Measurable beats: 31 of 34

Why measurement isn't possible:

Occurrences: 2. The QT end is unclear, or nearby RR intervals do not meet the correction conditions. See Measurement method for details.

Occurrences: 1. The waveform around this beat is incomplete, or part of the signal is unclear.

Correction can be less accurate at very fast or very slow heart rates. Interpretation also depends on age, sex and the formula used.

Measurement method: Uses QT from the same beat and the immediately preceding RR, both in seconds: Fridericia = QT / RR^(1/3); Bazett = QT / √RR. The algorithm requires continuous neighboring RR intervals, changes no greater than 10%, and intervals of 500–1500 ms. These are algorithm conditions, not a normal range.

TP segment

This segment runs from the end of one beat's T wave to the start of the next P wave. It is usually fairly flat and can provide a reference baseline for measurement.

TP-segment duration: Median 324 ms; min–max 297–340 ms; middle 80% 301–336 ms; Measurable beats: 32 of 34

Why measurement isn't possible:

Occurrences: 1. The waveform around this beat is incomplete, or part of the signal is unclear.

Occurrences: 1. This T-wave end or the next P-wave start is unclear, or there may be a signal gap between them.

At faster heart rates, this segment may be very short or hard to separate from adjacent waves.

Measurement method: Measured from this T-wave end to the next P-wave start. No value is produced if either boundary is unclear, the signal has a gap, or the interval is shorter than 40 ms.

RR / PP intervals

RR is the time between consecutive R peaks; PP is the time between consecutive P peaks. They reflect the intervals between electrical activity in the ventricles and atria, respectively.

RR interval: Median 883 ms; min–max 879–883 ms; middle 80% 880–883 ms; Measurable beats: 33 of 34

PP interval: Median 883 ms; min–max 879–883 ms; middle 80% 883–883 ms; Measurable beats: 32 of 34

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Why measurement isn't possible:

Occurrences: 1. The previous beat is missing, the P wave is unclear, or the signal has a gap between beats.

Occurrences: 1. The waveform around this beat is incomplete, or part of the signal is unclear.

These values describe variation in heartbeat intervals. Normal beats have not been separated from other types, so they cannot be treated as HRV.

Measurement method: RR and PP are calculated from their own consecutive peaks. They do not substitute for one another, and intervals are not joined across signal gaps.

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HeartPro — Local notes

Sep 18, 2026 at 4:08 PM; duration: 2 min
Caffeine
USERNOTE café 日本語 한글

Self-reported notes; for reference only, not for medical use